1) computer vision is a branch of Artificial intelligence that enables the computers to capture, process, analyze and interpret the to understand the images and videos

a) * To understand visual information
* To recognize the objects
* To automate the decision-making
* To improve accuracy and efficiency

b) Examples:
* Face recognizing systems
* self-driving cars.

46/50

2) computer vision extracts useful information from the images & videos and helps the machine to recognize the objects.

a) Feature extraction
Feature extraction is the process of identifying the important characteristics such as edges, shapes and colour from an image

b) object recognition identifies and classifies objects in images enabling applications like robotics & self driving cars.

3) a) low-level preprocessing improves the image quality and removes the noise, filtering and contrast enhancement

b)

Mid level	High Level
image segmentation & feature extraction	object recognition and scene understanding
produces image features	produces meaningful interpretation
intermediate processing	Final decision-making stage

4)
A digital image is represented as a matrix of pixels. Each pixel stores intensity or colour values

a) (Higher)
A pixel is the smallest unit of the digital image that rep -resents brightness or colour

b) High resolution means more pixels, resulting in better im -age quality and more detail. Lower resolution produces blury images.

5)
image acquisition captures light from a scene using an image sensor and converts it into a digital data

a) An image sensor converts light into electrical signals that are processed into a digital images

b) * Digital camera
* web cam

6)
a) sampling converts a continuous image into a discrete grid of pixels

b) quantization assigns numerical intensity values to pixels, enabling digital storage and processing.

7)
The image formation model explains how light from objects passes through the camera lens and forms an image on the sensor

a) perspective projection is the process of projecting 3D scene onto a 2D image plane, making distant object appear similar

b) illumination provides light needed to capture images proper lightning improves image quality and the object visibility

8) a)
image processing
-fying the imag
b)
Image processi
enhances im
output is an
improved im
focus on f

9) a)
computer
helping
b)
It de
self-

10) a)
pixe
-
fea
b)

8) a)
image processing is the technique of enhancing or modi-fying the images.

b)

Image processing	computer vision
enhances images	understands image content
output is an improved image	meaningful image
focus on pixels	object interpretation

9) a)
computer vision detects diseases from x-rays, CT scans helping doctors diagnose diseases early

b)
It detects lanes, traffic signals to enable safe self-driving

10) a) pixels contains the image information used for feature extraction, object detection and analysis

b) Higher sampling ~~and~~ higher preserve more image details leading to more accurate object detection.